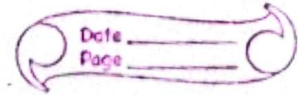


Tutorial 2



1. Find Z-transform using the properties:

a) $na^n u[n]$

Soln.

we know,

$$u[n] \xrightarrow{Z} \frac{1}{1-z^{-1}} \quad |z| > 1$$

$$a^n u[n] \xrightarrow{Z} \frac{1}{1-az^{-1}} \quad |z| > a$$

and,

$$\begin{aligned} na^n u[n] &\xrightarrow{Z} -z \frac{d}{dz} \left(\frac{1}{1-az^{-1}} \right) \\ &= -z (-1) (1-az^{-1})^{-2} \times -a(-1) z^{-2} \\ &= \frac{az^{-1}}{(1-az^{-1})^2} \quad |z| > a \end{aligned}$$

b) $a^n u[n-2]$

we know,

$$u[n] \xrightarrow{Z} \frac{1}{1-z^{-1}} \quad |z| > 1$$

using time shifting property,

$$u[n-2] \xrightarrow{Z} z^{-2} \times \frac{1}{1-z^{-1}} \quad |z| > 1$$

$$= \frac{z^{-2}}{1-z^{-1}} \quad |z| > 1$$

$$a^n u[n-2] \xrightarrow{Z} = \frac{(a^{-1}z)^{-2}}{1-(a^{-1}z)^{-1}} \quad |z| > 1$$

$$= \frac{4z^{-2}}{1-2z^{-1}} \times \frac{z}{z}$$

$$= \frac{4z^{-1}}{z-2}$$

c) $n^2 a^n u[n]$
 $\mathcal{Z}(a^n u[n]) = \frac{1}{1-az^{-1}} \quad |z| > |a|$

$$\begin{aligned} n^2 a^n u[n] &= z^2 \frac{d}{dz} \left(\frac{1}{1-az^{-1}} \right) \\ &= z^2 \cdot (-1)(1-az^{-1})^{-2} \times (-a)(-1)z^{-2} \\ &= \frac{-a}{(1-az^{-1})^2} \end{aligned}$$

d) $3\left(-\frac{1}{2}\right)^n u[n] - 2(3)^n u[-n-1]$

Soln.

$$X(z) = 3X_1(z) - 2X_2(z)$$

Then,

$$X_1(z) = \mathcal{Z}\left(\left(-\frac{1}{2}\right)^n u[n]\right)$$

$$= \frac{1}{1 + \frac{1}{2}z^{-1}} \quad |z| > \left|-\frac{1}{2}\right|$$

$$X_2(z) = \mathcal{Z}(3^n u[-n-1])$$

$$= \frac{1}{1-3z^{-1}} \quad |z| < |3|$$

$$\therefore X(z) = 3 \times \frac{2}{2+z^{-1}} \quad |z| > \frac{1}{2} \quad - \quad 2 \times \frac{1}{1-3z^{-1}} \quad |z| < 3$$

2. Find the system function $H(z)$ and unit sample response $h[n]$ of the following system. $y[n] = \frac{1}{2} y[n-1] + 2x[n]$

Solⁿ.

Taking Z transform on both side.

$$Y(z) = \frac{1}{2} z^{-1} Y(z) + 2X(z)$$

$$Y(z) \left(1 - \frac{1}{2} z^{-1}\right) = 2X(z)$$

$$\frac{Y(z)}{X(z)} = \frac{2}{1 - \frac{1}{2} z^{-1}}$$

$$H(z) = \frac{2}{1 - \frac{1}{2} z^{-1}}$$

Taking inverse z transform.

$$h[n] = 2 \cdot \left(\frac{1}{2}\right)^n u[n]$$

3. Determine the response of the system $y[n] = \frac{5}{6} y[n-1] - \frac{1}{6} y[n-2] + x[n]$ to the input signal.

$$x[n] = \delta[n] - \frac{1}{3} \delta[n-1]$$

Solⁿ.

Taking z transform on both side of system.

$$Y(z) = \frac{5}{6} z^{-1} Y(z) - \frac{1}{6} z^{-2} Y(z) + X(z)$$

$$\frac{Y(z)}{X(z)} = \frac{1}{1 - \frac{5}{6} z^{-1} + \frac{1}{6} z^{-2}}$$

$$X(z) = 1 - \frac{1}{3} z^{-1}$$

$$H(z) = \frac{1}{1 - \frac{5}{6} z^{-1} + \frac{1}{6} z^{-2}}$$

$$Y(z) = X(z) \times H(z)$$

$$= \frac{1 - \frac{1}{3}z^{-1}}{1 - \frac{5}{6}z^{-1} + \frac{1}{6}z^{-2}} \times 1$$

$$= \frac{1 - 0.33z^{-1}}{1 - 0.833z^{-1} + 0.666z^{-2}}$$

$$z = 0.33$$

$$p_1 = 0.416 + 0.701j = 0.815 \angle 59.9^\circ$$

$$p_2^* = 0.416 - 0.701j = 0.815 \angle -59.9^\circ$$

$$Y(z) = \frac{A}{(1 - pz^{-1})} + \frac{A^*}{(1 - p^*z^{-1})}$$

$$\begin{aligned} A &= \left. \frac{1 - 0.33z^{-1}}{1 - p^*z^{-1}} \right|_{z=p} = \frac{1 - 0.33p^{-1}}{1 - p^*p^{-1}} \\ &= \frac{1 - 0.33}{0.416 + 0.701j} \\ &= \frac{0.67}{1 - (0.416 - 0.701j)} \\ &= \frac{0.67}{0.416 + 0.701j} \\ &= 0.5 - 0.061j \\ &= 0.503 e^{-j6.99} \end{aligned}$$

$$A^* = 0.503 e^{j6.99}$$

$$Y(z) = \frac{0.503 e^{-j6.99}}{1 - pz^{-1}} + \frac{0.503 e^{j6.99}}{1 - p^*z^{-1}}$$

$$\begin{aligned} y[n] &= 0.503 e^{-j6.99} (p^n) u[n] + 0.503 e^{j6.99} (p^*)^n u[n] \\ &= 0.503 e^{-j6.99} 0.815^n e^{j59.31n} u[n] + 0.503 e^{j6.99} 0.815^n e^{-j59.31n} u[n] \\ &= 0.503 \cdot 0.815^n (e^{j(59.31n - 6.99)} + e^{-j(59.31n - 6.99)}) u[n] \\ &= 0.503 \cdot 0.815^n 2 \cos(59.31n - 6.99) u[n] \end{aligned}$$

4. Find inverse Z transform.

a) $X(z) = \frac{z^3}{(z+1)(z-1)^2}$, find causal signal.

Soln.

$$X(z) = \frac{A}{z+1} + \frac{B}{z-1} + \frac{C}{(z-1)^2}$$

$$A = \left. \frac{z^3}{(z-1)^2} \right|_{z=-1} = \frac{-1}{4}$$

$$C = \left. \frac{z^3}{z+1} \right|_{z=1} = \frac{1}{2}$$

$$\begin{aligned} B &= \left. \frac{d}{dz} \left(\frac{z^3}{z+1} \right) \right|_{z=1} \\ &= \left. \frac{(z+1)3z^2 - z^3 \cdot 1}{(z+1)^2} \right|_{z=1} \\ &= \frac{6-1}{4} = \frac{5}{4} \end{aligned}$$

$$X(z) = \frac{-\frac{1}{4} \cdot 1}{1-(-z)} + \frac{5}{4}$$

$$= -\frac{1}{4} \frac{z^{-1}}{1+z^{-1}} + \frac{1}{2} \frac{z^{-1}}{1-z^{-1}} + \frac{5}{4} \frac{z^{-2}}{(1-z^{-1})^2}$$

$$= -\frac{1}{4} (-1)^n u[n-1] + \frac{1}{2} u[n] + \frac{5}{4} n u[n-2]$$

Q6. $X(z) = \frac{z}{3z^2 - 4z + 1}$, find anticausal signal,

$$= \frac{z^{-1}}{3 - 4z^{-1} + z^{-2}}$$

$$= \frac{z^{-1}}{(1 - z^{-1})(1 - 0.33z^{-1})}$$

$$X(z) = \frac{A}{1 - z^{-1}} + \frac{B}{1 - 0.33z^{-1}}$$

$$A = \frac{z^{-1}}{1 - 0.33z^{-1}} \Big|_{z=1} = 1.492$$

$$B = \frac{z^{-1}}{1 - z^{-1}} \Big|_{z=0.33} = -1.492$$

$$X(z) = \frac{1.492}{1 - z^{-1}} - \frac{1.492}{1 - 0.33z^{-1}}$$

$$= 1.492 - u[-n-1] + 1.492 (0.33)^n u[-n-1]$$

5. Determine the step response of the causal system $y[n] = y[n-1] + n[n]$. Is this stable system?
Solⁿ.

Since input is step response so replace $n[n]$ by $u[n]$

$$y[n] = y[n-1] + u[n]$$

$$Y(z) = z^{-1}Y(z) + \frac{1}{1-z^{-1}}$$

$$Y(z) = \frac{1}{1-z^{-1}} \cdot \frac{1}{1-z^{-1}}$$

$$Y(z) = \frac{1}{(1-z^{-1})^2}$$

$$\mathcal{Z}^{-1} \left\{ \frac{1}{(1-z^{-1})^2} \right\} \xrightarrow{(n+1)u[n+1]} \frac{z \cdot z^{-1}}{(1-z^{-1})^2}$$

$$y[n] = nu[n] \quad y[n] = n+1$$

Since input n is independent in time domain so the system is unstable.

6. A LTI system is described by the difference equation
 $y[n] = x[n] + 2x[n-1] + x[n-2]$

a. Determine the impulse response of the system.

- The impulse response of the system $h[n]$ is output when the input is discrete time unit impulse function.

The unit impulse function is defined as:

$$\delta[n] = \begin{cases} 1, & n=0 \\ 0, & n \neq 0 \end{cases}$$

Substituting $x[n] = \delta[n]$

$$h[n] = \delta[n] + 2\delta[n-1] + \delta[n-2]$$

The impulse response is

$$h[n] = \begin{cases} 1, & n=0 \\ 2, & n=1 \\ 1, & n=2 \\ 0, & \text{otherwise} \end{cases}$$

b. Is this stable system?

Yes this system is stable because sum of all ^{impulse} response is finite. i.e. $|1| + |2| + |1| = 4$ which is finite.

c. Determine frequency response of the system i.e. $H(e^{j\omega})$

Soln.

$$H(e^{j\omega}) = 1 + 2e^{-j\omega} + e^{-2j\omega}$$

$$= 1 + e^{-j\omega} + e^{-j\omega} + e^{-2j\omega}$$

$$= 1 + e^{-j\omega} + e^{-j\omega}(1 + e^{-j\omega})$$

$$= 1 + e^{-j\omega}$$

$$= 1 + e^{-j\omega}(2 + e^{-j\omega})$$

7. When the input to an LTI system is $x[n] = (\frac{1}{2})^n u[n] + 2^n u[-n-1]$; the output is found to be $y[n] = 6(\frac{1}{2})^n u[n] - 6(\frac{3}{4})^n u[n]$.
- a. Find transfer function $H(z)$ and plot pole zero.

$$X(z) = \frac{1}{1 - \frac{1}{2}z^{-1}} \quad |z| > \frac{1}{2} + \frac{1}{1 - 2z^{-1}} \quad |z| < 2$$

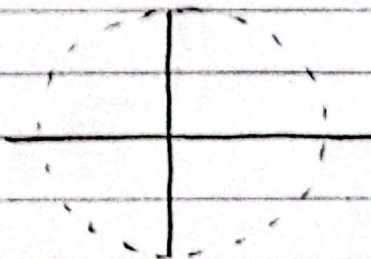
$$= \frac{1}{1 - \frac{1}{2}z^{-1}} + \frac{1}{1 - 2z^{-1}} \quad \frac{1}{2} < |z| < 2$$

$$Y(z) = 6 \frac{1}{1 - \frac{1}{2}z^{-1}} - 6 \frac{1}{1 - \frac{3}{4}z^{-1}} \quad |z| > \frac{3}{4}$$

$$\begin{aligned} H(z) &= \frac{Y(z)}{X(z)} = \frac{6 - 18/4 z^{-1} - 6 + 3z^{-1}}{(1 - \frac{1}{2}z^{-1})(1 - \frac{3}{4}z^{-1})} \\ &= \frac{1 - 2z^{-1} + 1 - \frac{1}{2}z^{-1}}{(1 - \frac{1}{2}z^{-1})(1 - 2z^{-1})} \\ &= \frac{-6/4 z^{-1} + 2 - 5/2 z^{-1}}{(1 - 2z^{-1})(1 - \frac{3}{4}z^{-1})} \\ &= \frac{2 - 4z^{-1}}{(1 - 2z^{-1})(1 - \frac{3}{4}z^{-1})} \\ &= \frac{2(1 - 2z^{-1})}{(1 - 2z^{-1})(1 - \frac{3}{4}z^{-1})} \end{aligned}$$

$$p_1 = 2 \quad p_2 = 3/4$$

$$z_1 = 2$$



6. Find impulse response.

$$U(z) = \frac{1}{1 - 3/4 z^{-1}}$$

Using inverse z transform.

$$\mathcal{Z}^{-1}\left\{\frac{1}{1 - 3/4 z^{-1}}\right\} = \frac{1}{1 - 3/4 z^{-1}}$$

$$\therefore h[n] = \mathcal{Z}^{-1}\left\{\frac{1}{1 - 3/4 z^{-1}}\right\}$$

7. Write the difference equation that characterizes the system.

$$y[n] = -\sum_{k=1}^N a_k y[n-k] + \sum_{k=0}^M b_k x[n-k]$$

8. Is the system stable? Is it causal?

b. Find impulse response.

$$H(z) = \frac{2}{1 - 3/4 z^{-1}}$$

Taking inverse z transform.

$$2 \cdot \left(\frac{3}{4}\right)^n u[n] \xrightarrow{z} \frac{2}{1 - 3/4 z^{-1}}$$

$$\therefore h[n] = 2 \cdot \left(\frac{3}{4}\right)^n u[n]$$

c. Write LCCD equation that characterize the system.

$$y[n] = -\sum_{k=1}^N a_k y[n-k] + \sum_{k=0}^M b_k x[n-k]$$

d. Is the system stable? Is it causal?

8. Consider the LTI system with transfer function $H(z) = \frac{z^{-2}}{(1-\frac{1}{2}z^{-1})(1-3z^{-1})}$

a. Suppose that the system is known to be stable, determine the output when the input is unit step.

Sol.

Since input is unit step so $x[n] = u[n]$.

$$X(z) = \frac{1}{1-z^{-1}} \quad |z| > 1$$

And,

$$\begin{aligned} Y(z) &= H(z) \times X(z) \\ &= \frac{z^{-2}}{(1-\frac{1}{2}z^{-1})(1-3z^{-1})} \times \frac{1}{1-z^{-1}} \\ &= \frac{z^{-2}}{(1-z^{-1})(1-\frac{1}{2}z^{-1})(1-3z^{-1})} \end{aligned}$$

Now,

$$Y(z) = \frac{A}{1-z^{-1}} + \frac{B}{1-\frac{1}{2}z^{-1}} + \frac{C}{1-3z^{-1}}$$

And,

$$A = \frac{z^{-2}}{(1-\frac{1}{2}z^{-1})(1-3z^{-1})} \Big|_{z=1} = -1$$

$$B = \frac{z^{-2}}{(1-z^{-1})(1-3z^{-1})} \Big|_{z=\frac{1}{2}} = \frac{4}{5}$$

$$C = \frac{z^{-2}}{(1-z^{-1})(1-\frac{1}{2}z^{-1})} \Big|_{z=3} = \frac{1}{5}$$

$$Y(z) = \frac{-1}{1-z^{-1}} + \frac{4/5}{1-\frac{1}{2}z^{-1}} + \frac{1/5}{1-3z^{-1}}$$

$$y[n] = -u[n] + \frac{4}{5}\left(\frac{1}{2}\right)^n u[n] + \frac{1}{5}(3)^n u[n]$$